



Islamic University of Technology

Department of Computer Science and Engineering (CSE)

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Final Project Report

TypeDrift

Group 2

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1 Problem Statement

Typing proficiency has become an essential skill in today's digital landscape, yet many individuals continue to struggle with inefficient typing habits that hinder productivity in both academic and professional settings. While a range of typing practice tools exists, they consistently fall short in several key areas:

- **High Latency and Bloat:** Web-based applications frequently suffer from input lag and cluttered interfaces, creating unnecessary distractions that disrupt focused practice.
- **Lack of Offline Accessibility:** Most platforms require a persistent internet connection to track progress or access advanced exercises, rendering them unsuitable for offline or low-resource environments.
- **Environment Mismatch:** Software developers, system administrators, and terminal power users lack tools that integrate naturally into their local command-line workflow. Existing solutions are often built around graphical interfaces that feel out of place for this audience.

2 Potential Solution

To address these shortcomings, we developed **TypeDrift**, a high-performance, lightweight Command-Line Interface (CLI) typing monitor built entirely in C++. TypeDrift operates completely offline, providing a distraction-free terminal environment with real-time performance metrics, comprehensive session analytics, and a secure local profile system. By keeping all data stored on the user's own machine and employing a minimal Text User Interface (TUI), TypeDrift eliminates input latency and ensures full data privacy. The application is designed to feel at home in the same environment where developers spend most of their time: the terminal.

3 Implemented Features

The following features were successfully implemented in the final build, based on the original project proposal and refinements made during development:

- **Real-Time Performance Engine**

- Continuously calculates Words Per Minute (WPM) and accuracy throughout each session.
- Delivers live performance feedback without interrupting the typing flow.

- **Color-Coded Visual Feedback**

- Untyped characters are rendered in grey to indicate pending input.
- Correct keystrokes are highlighted in green; incorrect ones are marked in red.
- Provides clear, at-a-glance feedback with no reliance on additional UI elements.

- **Secure Local Authentication**

- Maintains a persistent user profile system across sessions.
- Employs custom cryptographic password hashing to protect stored credentials.

- **Performance History and Data Persistence**

- Session results are saved locally after every completed run.
- Users can revisit past scores to monitor improvement over time.

- **Dynamic Word Engine**

- Generates randomized text passages across sessions to prevent repetition.
- Ensures each practice run presents a genuinely new challenge.

- **Difficulty Tiers and Game Modes**

- Offers multiple pre-configured difficulty levels suited to different skill ranges.

- Includes additional challenge modes to maintain engagement and variety.
- **Local Leaderboard**
 - High scores are stored persistently on the local machine.
 - Supports healthy competition among multiple users sharing the same system.
- **Unix-Based CLI Engine**
 - Optimized for terminal environments on Linux and macOS.
 - Leverages Unix system capabilities for responsive, low-latency input handling.

4 Tools and Technology

The project was developed with an emphasis on modern software engineering practices, solid Object-Oriented Programming (OOP) principles, and minimal external dependencies. The tools and technologies used are listed below.

Table 1: Tools and Technologies Used

Category	Tool / Technology
Core Programming Language	C++ (C++17 Standard)
Build System	CMake & Makefile
Integrated Development Environment	Visual Studio Code (VS Code)
Version Control	Git
Collaboration & Hosting	GitHub
Target Operating System	Linux and macOS (Unix-based)

5 Application Domain

TypeDrift falls under the domain of **Educational Software** and **Productivity Tools**, designed specifically for people who live in the terminal. Its primary audience includes software developers, system administrators, students, and keyboard enthusiasts who spend a significant portion of their day in CLI environments and want to sharpen their typing without ever leaving one.

Beyond raw speed, TypeDrift addresses a practical need: the gap between knowing how to type and typing well under pressure. It treats the terminal not just as a delivery mechanism, but as the natural habitat of its users, making the practice environment identical to the one they actually work in. This makes improvement feel less like a drill and more like a natural extension of the workflow.

6 Project Timeline

The project was carried out over the course of the semester. The Gantt chart below illustrates the major development milestones and their distribution across the project duration.

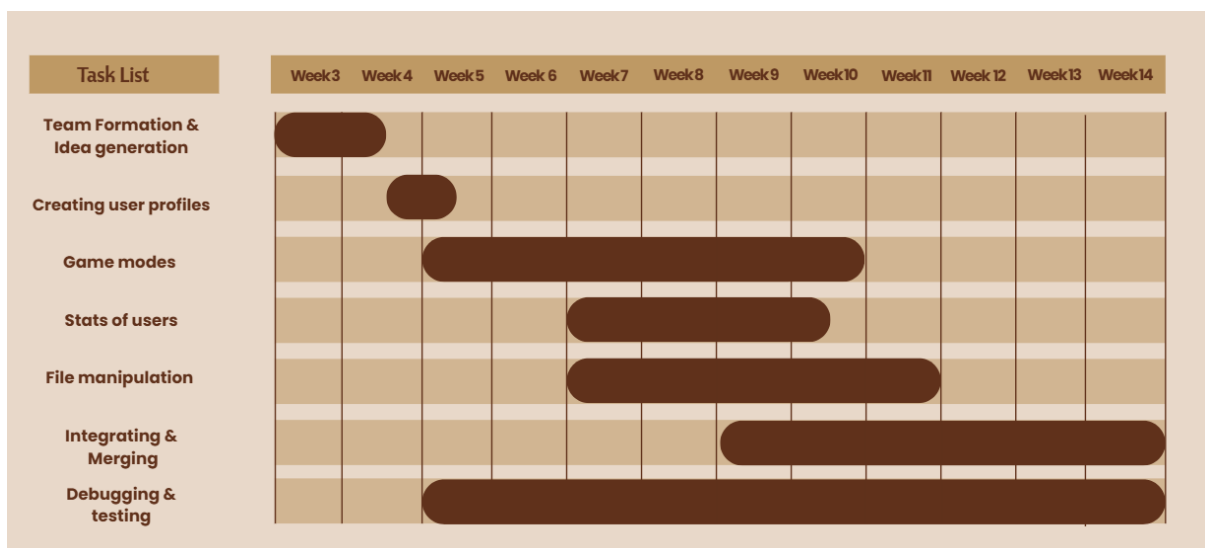


Figure 1: Project development timeline (Weeks 3–14)

7 Contribution

The project was a collaborative effort, with responsibilities divided according to each member's strengths and the natural boundaries of the system's components.

- **S. M. Samiul Hossain [230042110]**

- Designed the core game loop and overall TUI layout.
- Implemented real-time WPM and accuracy tracking algorithms.
- Built the dynamic text randomization engine.
- Integrated the CMake build system.
- Implemented the Time Control game mode.

- **Md. Ashraful Abedin [230042111]**

- Developed the secure user authentication system.
- Implemented custom password hashing and the user registration flow.
- Contributed to data encryption and credential security.
- Implemented the No-Vowel game mode.

- **Shaikh Muhaiminul Hasan [230042113]**

- Designed and implemented the core file handling system.
- Built the multi-user profile management and session history storage.
- Developed the local leaderboard with persistent score tracking.
- Implemented the Falling Words game mode.

8 Future Work

While TypeDrift meets its core objectives, several meaningful enhancements could be pursued in future development cycles:

- **Global Leaderboards:** Transitioning the current local leaderboard to a cloud-based system would enable competitive ranking among users across different machines worldwide.
- **Advanced Analytics:** Implementing terminal-based visual charts—such as ASCII or Unicode graphs—to display WPM and accuracy trends over time, giving users a clearer picture of their long-term progress.
- **Adaptive Learning Engine:** Introduce an intelligent “Custom Mode” that uses machine learning to analyze user errors and generate targeted practice sessions.
- **Customizable Themes:** Giving users the ability to personalize CLI colors and text styles to better suit their preferences and terminal setup.
- **Multiplayer Racing Mode:** A socket-based local network feature that would allow two or more users to compete against each other in real-time typing races directly within the terminal.
- **Remote Synchronization:** An optional cloud sync feature to back up and share progress across multiple devices.

9 Source Repository

The complete source code, documentation, and build instructions for TypeDrift are publicly available on GitHub:

GitHub URL: <https://github.com/AshrafulAbedin/TypeDrift>